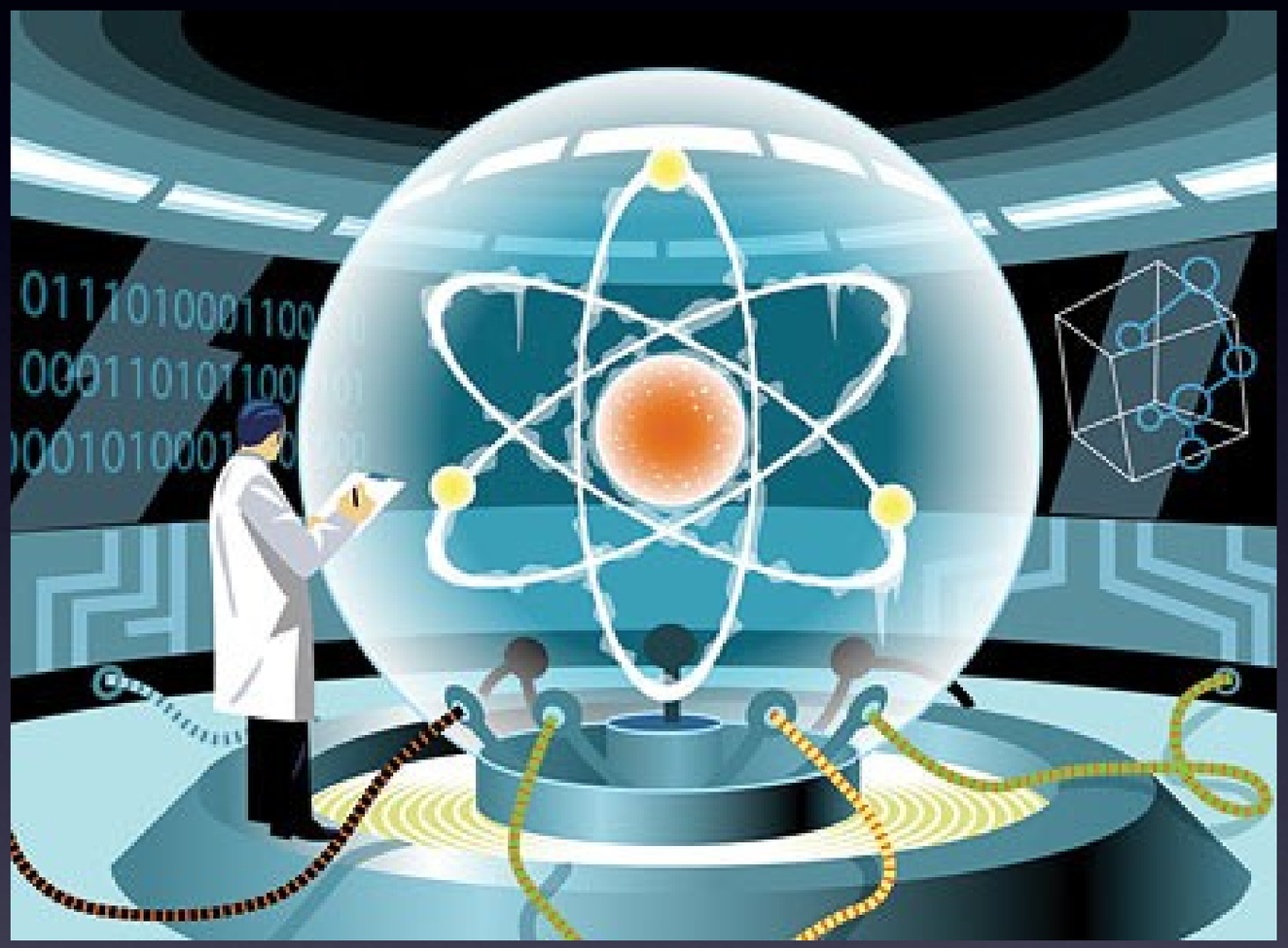
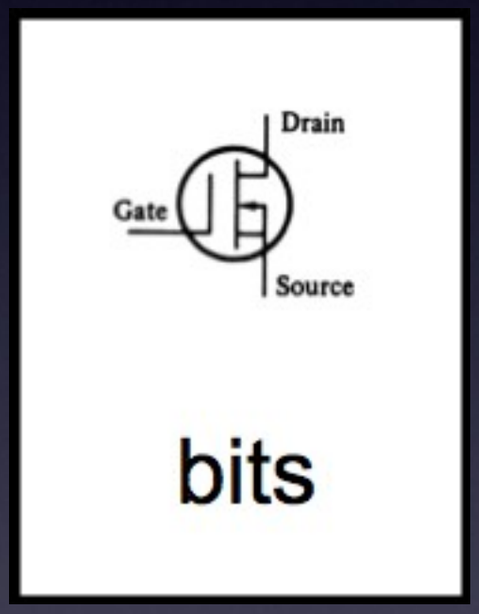
Quantum Computing



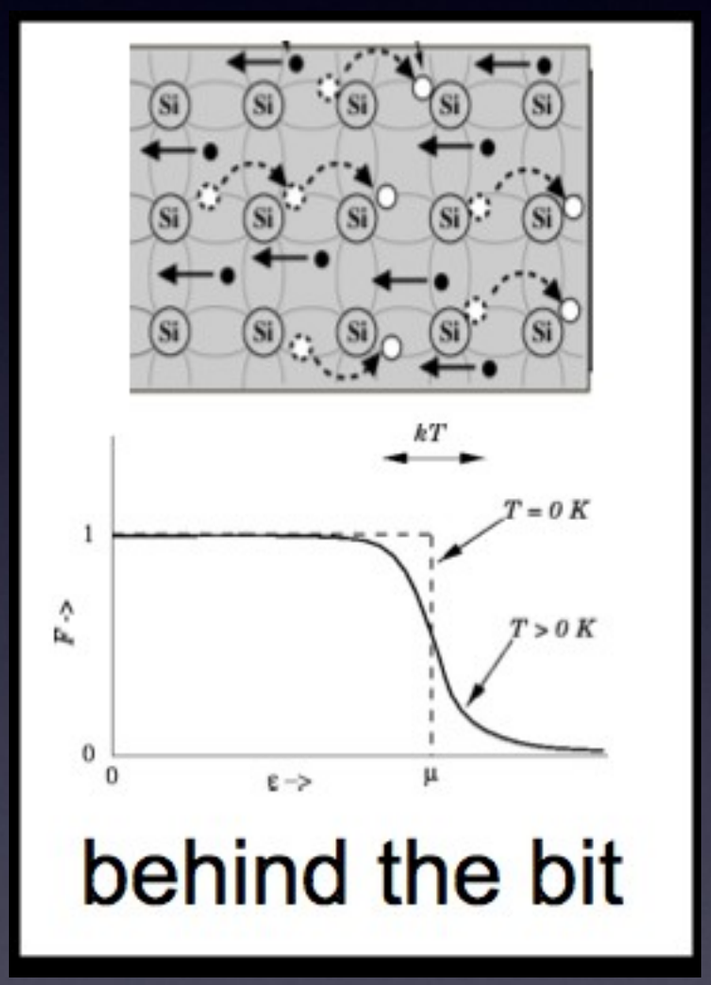
Dave Bacon
University of Washington

A Simple Question

Why is computation physically possible?



bits



behind the bit

Category Errors

Analog / Digital



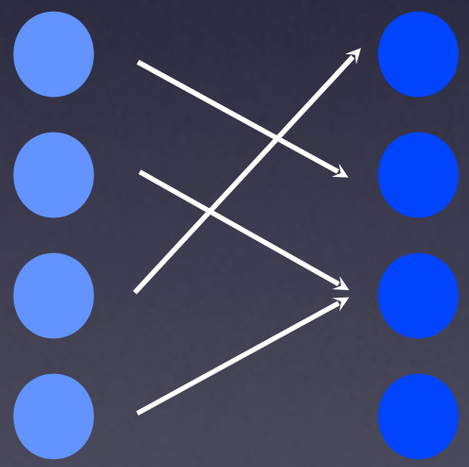
vs



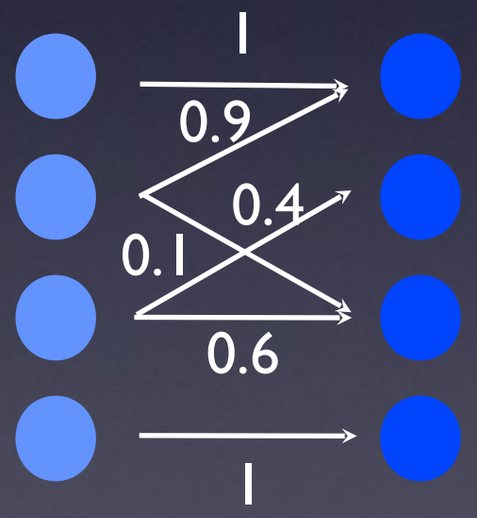
0.99821312 3....

0

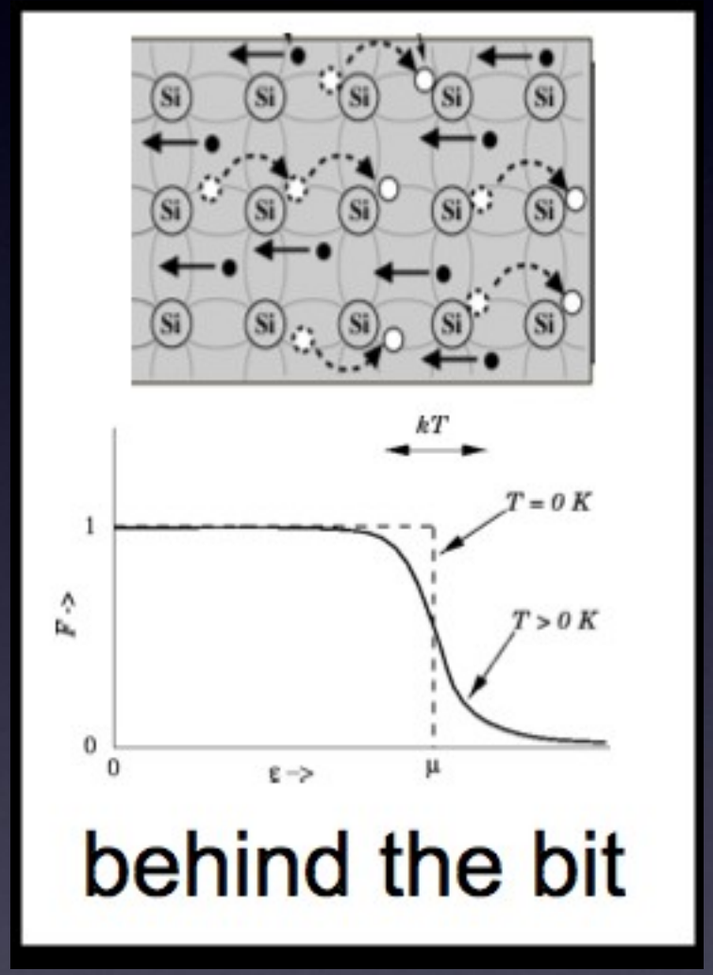
Deterministic / Probabilistic



vs

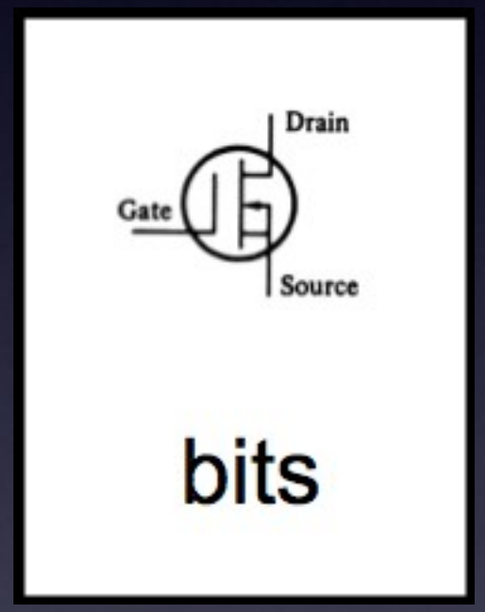


Analog Probabilistic
Digital Deterministic



Analog
Voltages

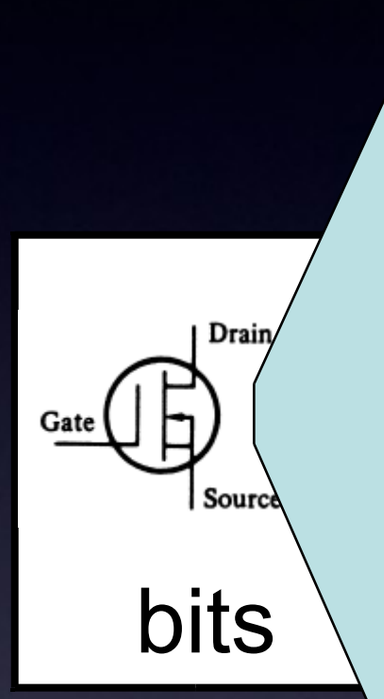
Digital
Signal



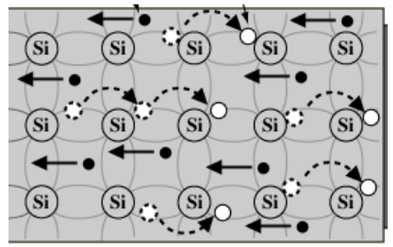
Probabilistic
Electrons

Deterministic
Gates

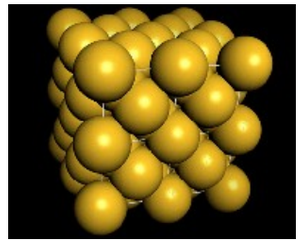
Deeper

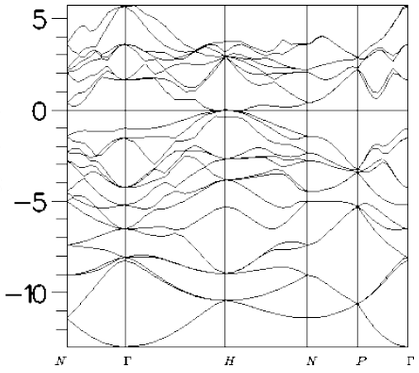


bits



noisy bits

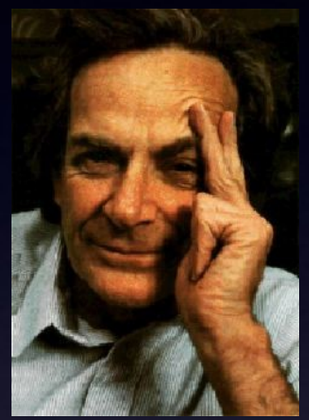




quantum bit

Deterministic / Probabilistic / Quantum

Quantum Propaganda



I think I can safely say that nobody understands quantum mechanics.

Richard Feynman
Nobel Prize 1965



Anyone who is not shocked by quantum theory has not understood it.

Niels Bohr
Nobel Prize 1922

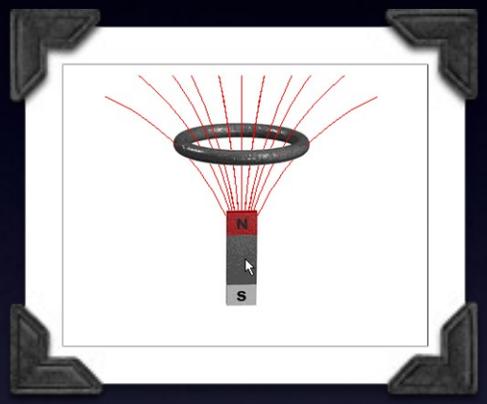
Hogwash



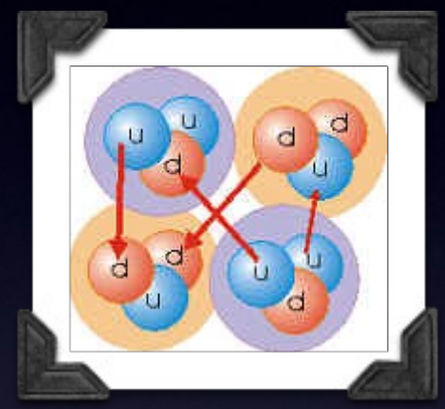
- They understand it
- They use it
- But do they believe it?

What is Quantum Theory?

Electromagnetism

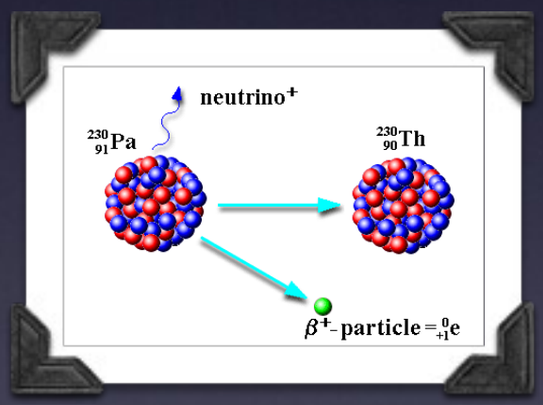


Strong force

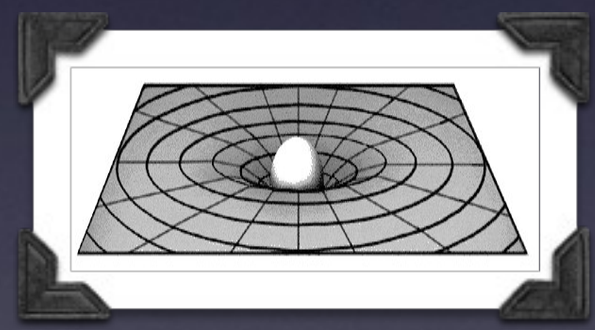




Weak force



Gravity ?!?!

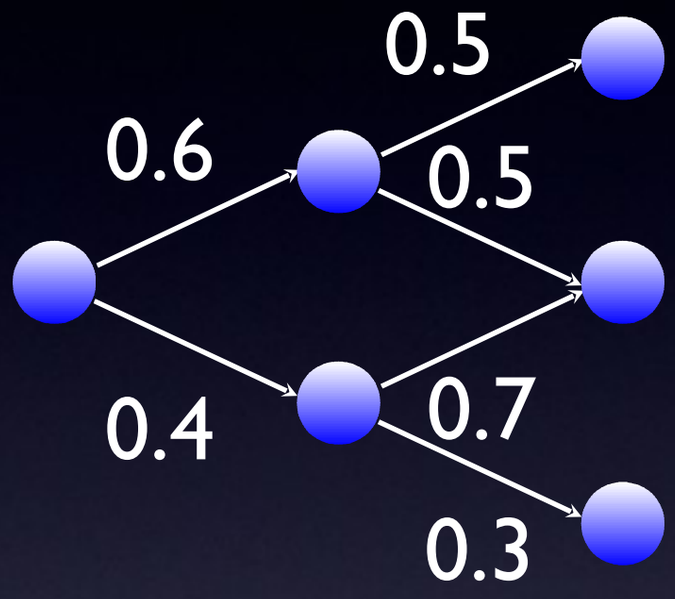


Quantum Theory

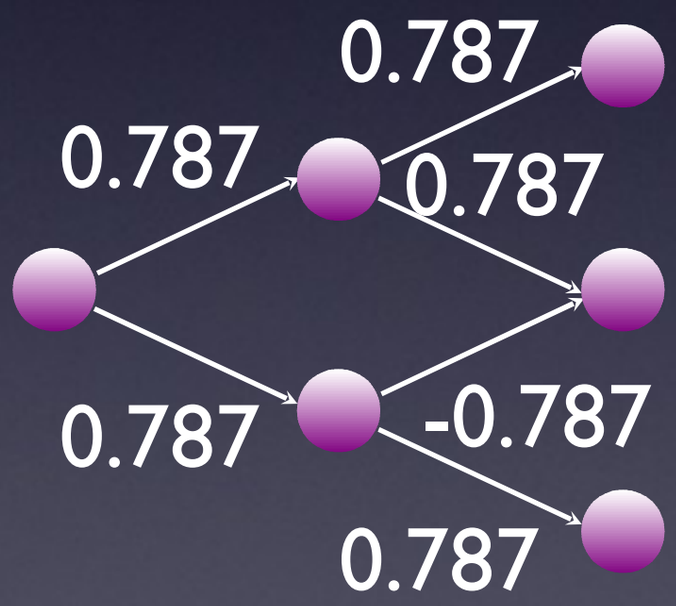
Bits, Pbits, Qubits

	bit	probabilistic bit	quantum bit
Configurations:	0 1	0 1	0 1
Description:	$\begin{bmatrix} 1 \\ 0 \end{bmatrix}$	$\begin{bmatrix} p \\ 1-p \end{bmatrix}$ $p \in \mathbb{R}$	$\begin{bmatrix} \alpha \\ \beta \end{bmatrix}$ $\alpha, \beta \in \mathbb{C}$
Observation:	0 certainty	0 p percent 1 1-p percent	0 $\lvert\alpha\rvert^2$percent 1 $\lvert\beta\rvert^2$percent
Evolution:	$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ deterministic	$\begin{bmatrix} 1-q & r \\ q & 1-r \end{bmatrix}$ stochastic probabilistic	$\begin{bmatrix} u & v \\ w & x \end{bmatrix}$ unitary

Garden of Re-forking Paths



$(0.6)(0.5)+(0.4)(0.7)=0.58$



$|(0.787)(0.787)+(-0.787)\ (0.787)|^2=0$

Quantum Computers

188198812920607963838697239461650439807163563379417382700763356422988859715234665485319060606504743045317388011303396716199692321205734031879550656996221305168759307650257059

=

x

Quantum Computers

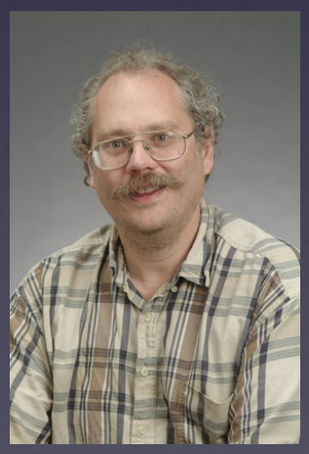
188198812920607963838697239461650439807163563379417382700763356422988859715234665485319060606504743045317388011303396716199692321205734031879550656996221305168759307650257059

=

398075086424064937397125500550386491199064362342526708406385189575946388957261768583317

x

472772146107435302536223071973048224632914695302097116459852171130520711256363590397527



Peter Shor, 1994:

- Quantum computers can efficiently factor
- No known classical efficient algorithm.
- Breaks RSA, Diffie-Hellman cryptosystems.

Stop Worrying and Love Quantum Theory

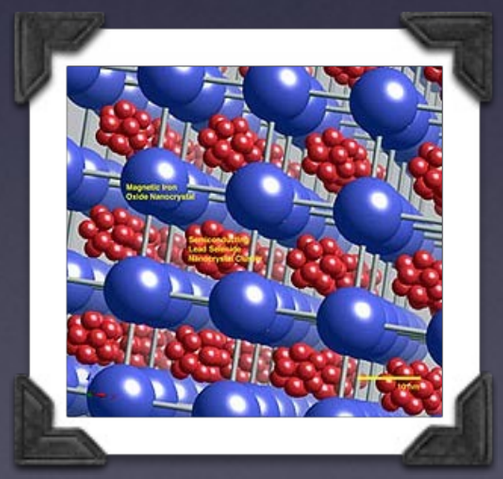


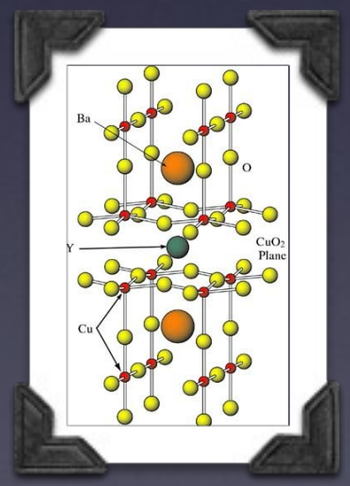
Quantum Algorithms

Exponential speedup

Shor's algorithm
- Breaks RSA, Diffie-Helman
- Only a few alternatives which are note broken by quantum computers

Simulation of quantum physics





Polynomial speedup

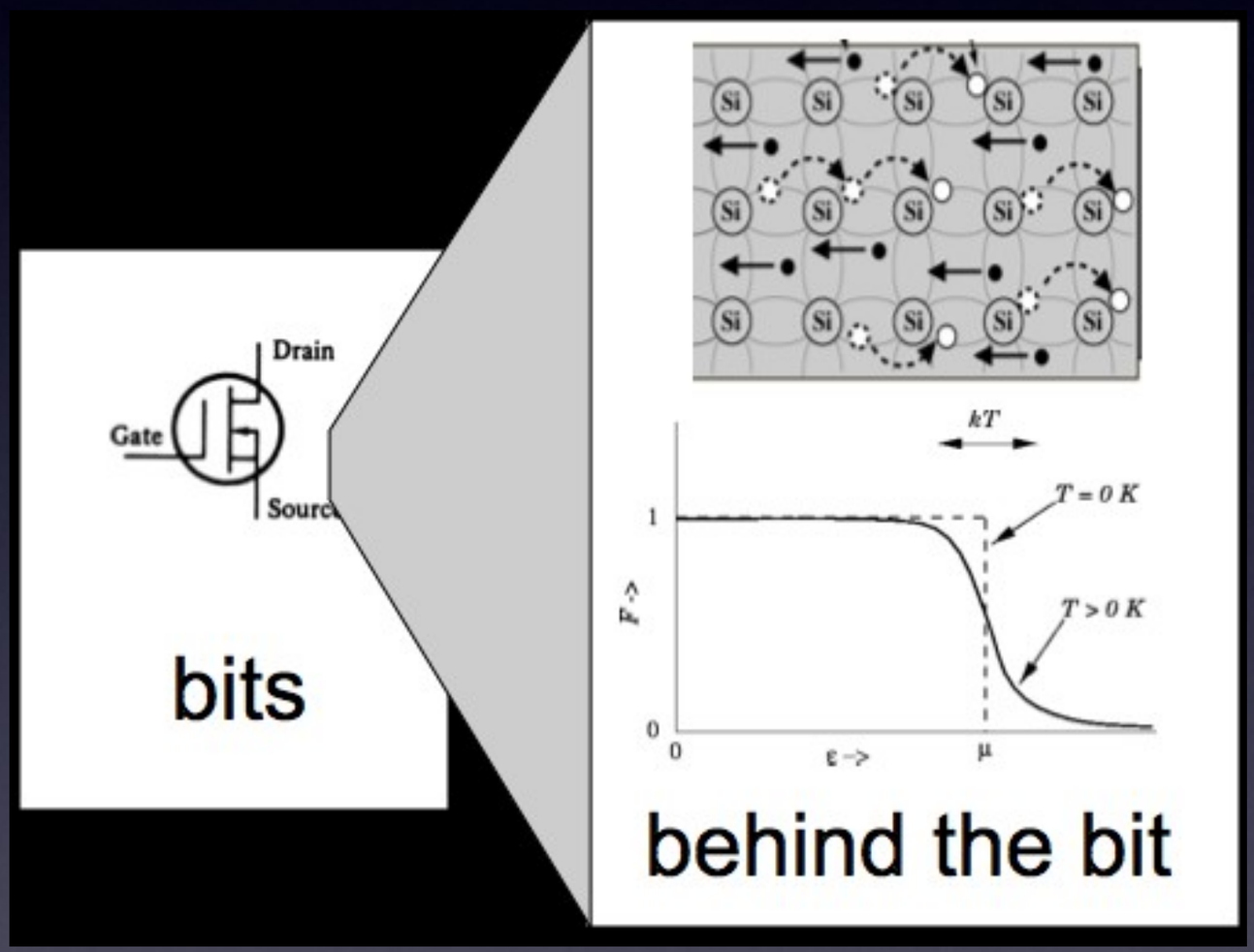
Grover's algorithm
- quadratic speedup in needle-in-haystack search

NAND tree algorithm
- quadratic speedup in game algorithms

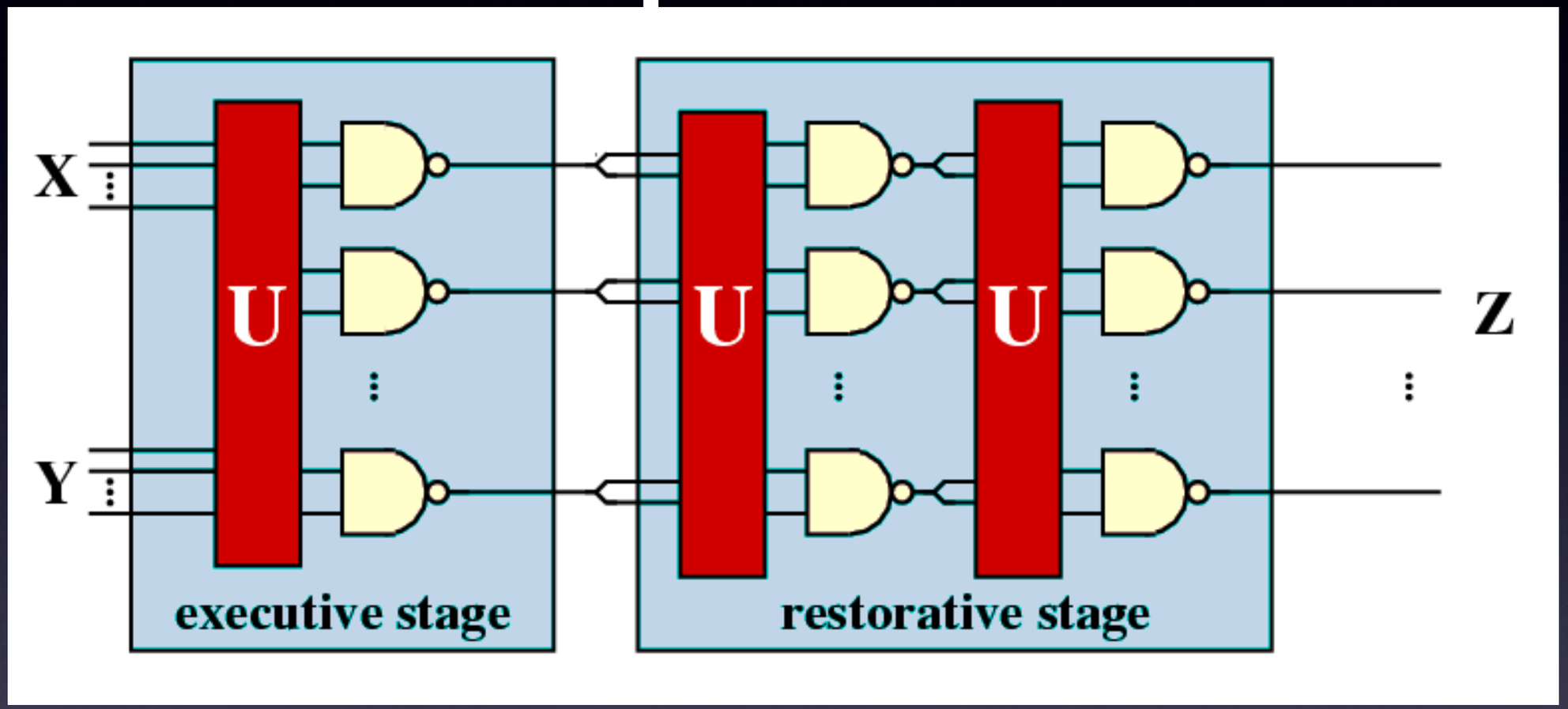
Amplitude amplifiction
- Ubiquitous quadratic speedups

A Simple Question

Why is computation physically possible?

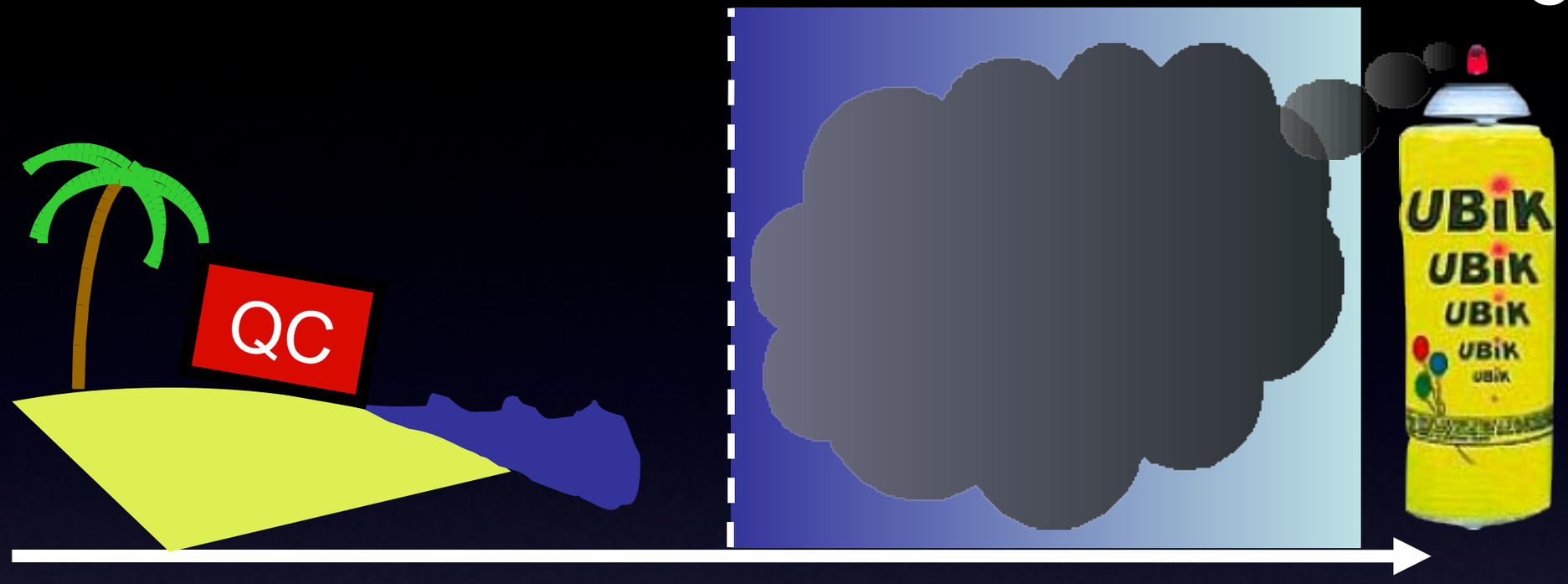


Fault-Tolerant Computation



von Neumann 1952

Threshold Theorem, 1996



“decoherence”

“imprecision”

Threshold Theorem for Fault-Tolerant Quantum Computation

[Aharonov, Ben-Or, Knill, LaFlamme, Zurek, Shor, Gottesman, Preskill]

If noise is weak enough and control is precise enough, then a quantum computation to a desired accuracy can be carried out with an overhead polynomial in the log of the accuracy.

Architectures For Quantum Computers

Systems-level challenges to achieving scalable, fault-tolerant quantum computation



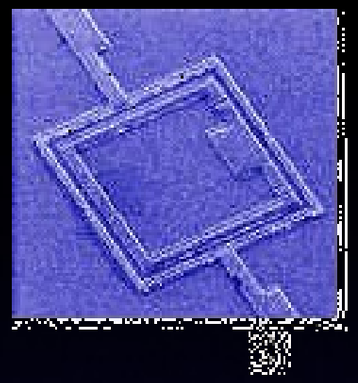
Error Correction

Designing computer architectures when components fail

The error correcting architectures must be massively (but fairly embarrassingly) parallel

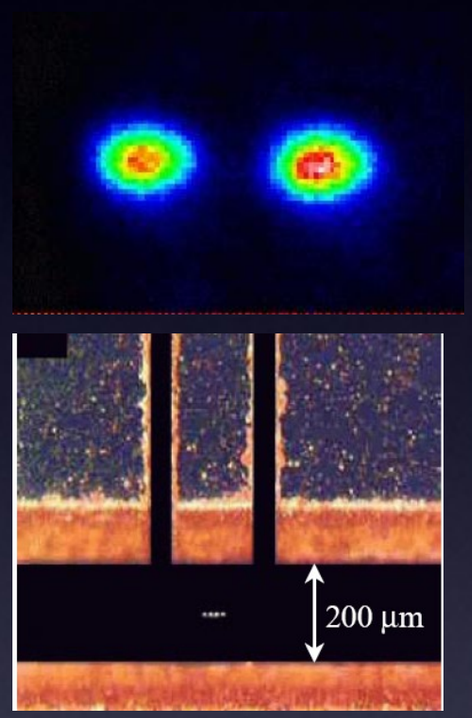
Building Blocks

stationary qubits



Example: superconducting qubits

walking "slow"



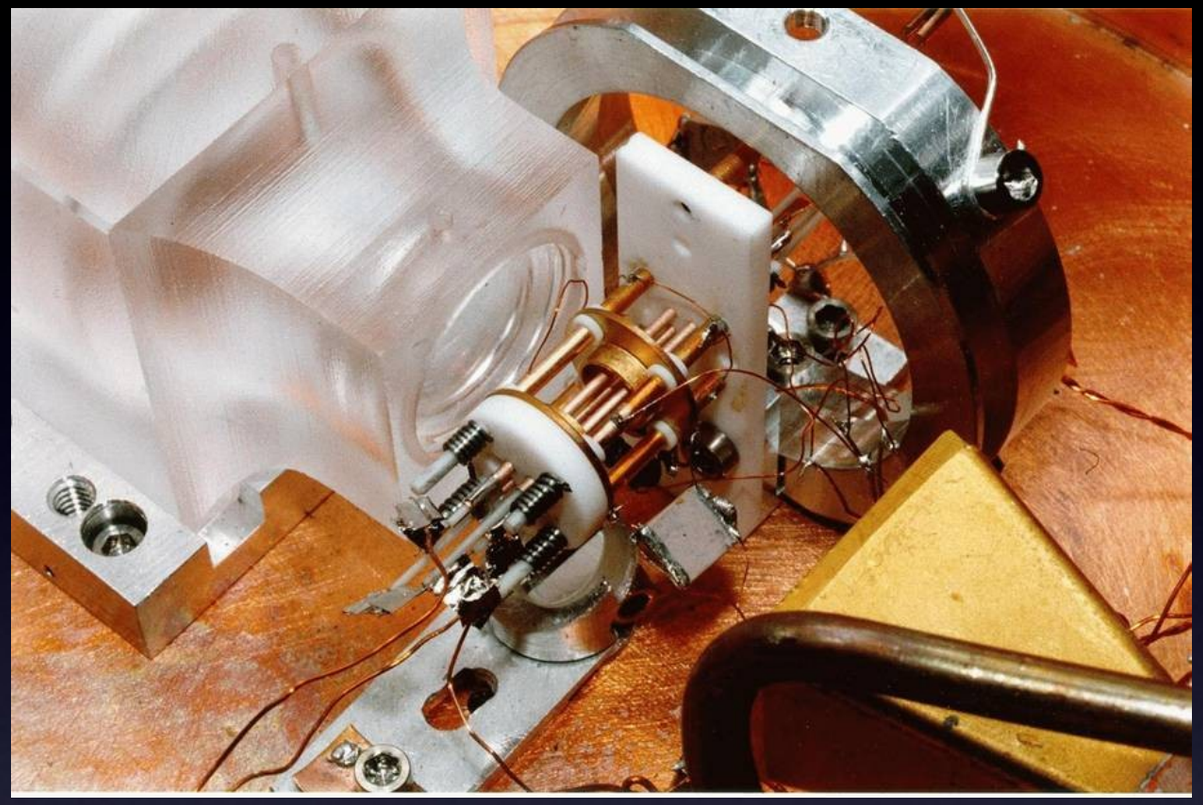
Example: ion trap qubits

flying "fast"



Example: photons

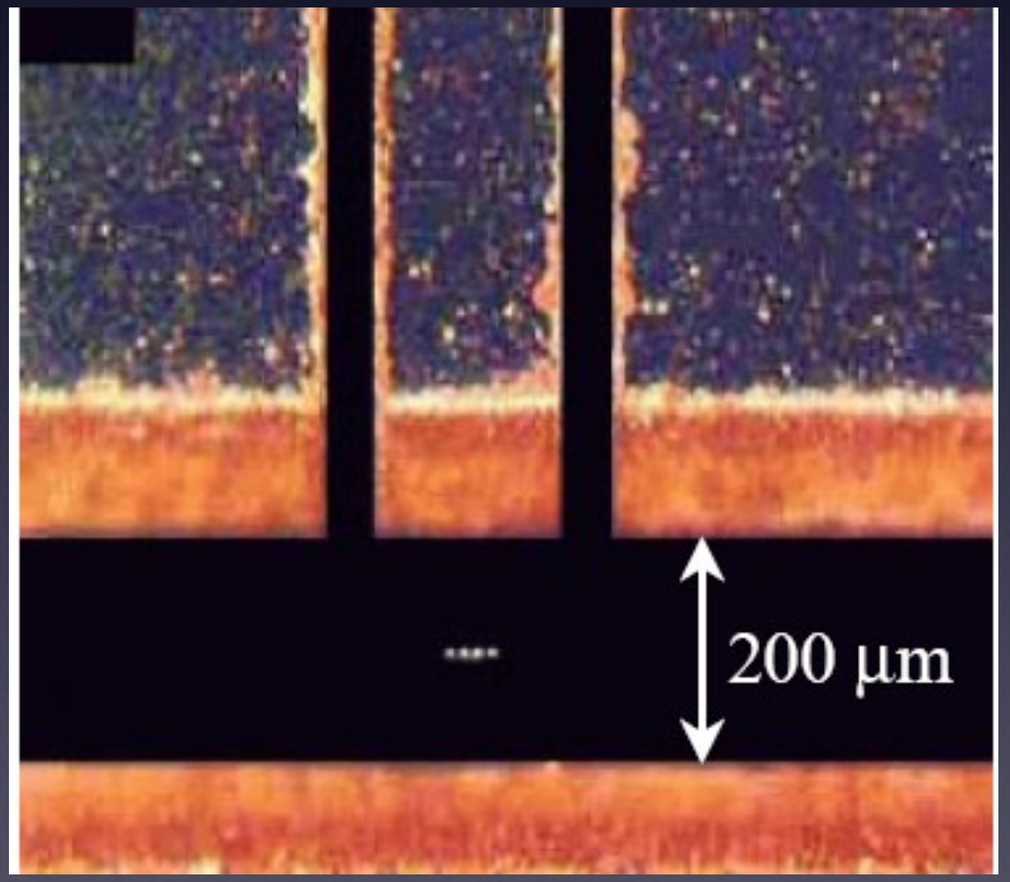
Ion Trap

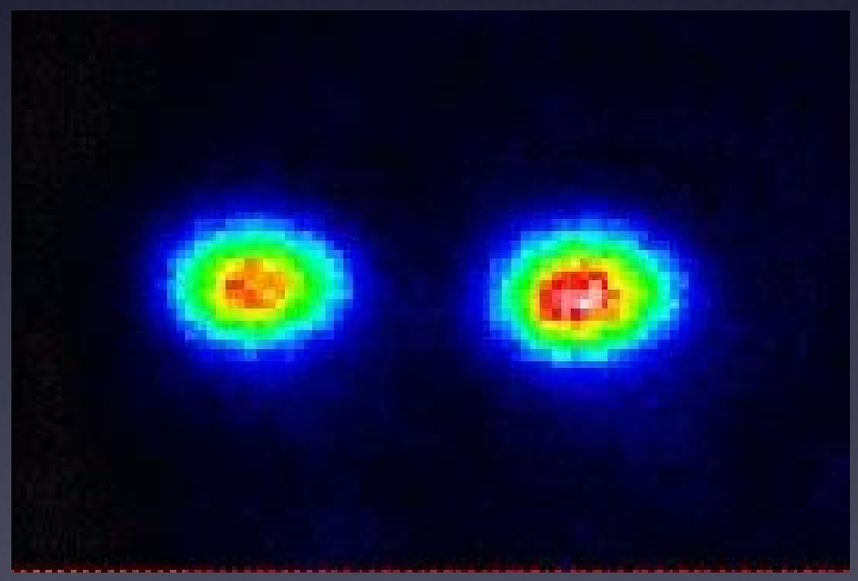


Oscillating electric fields trap ions

like charges repel

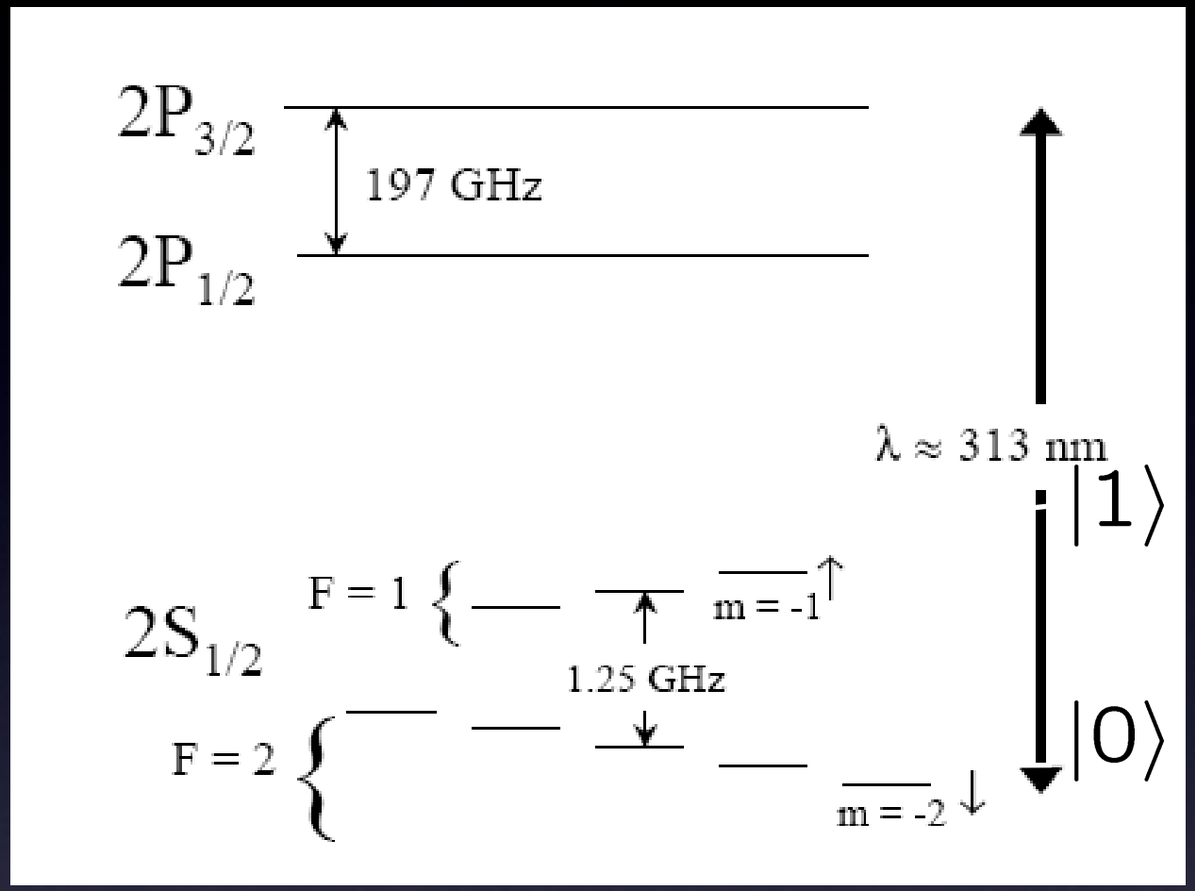


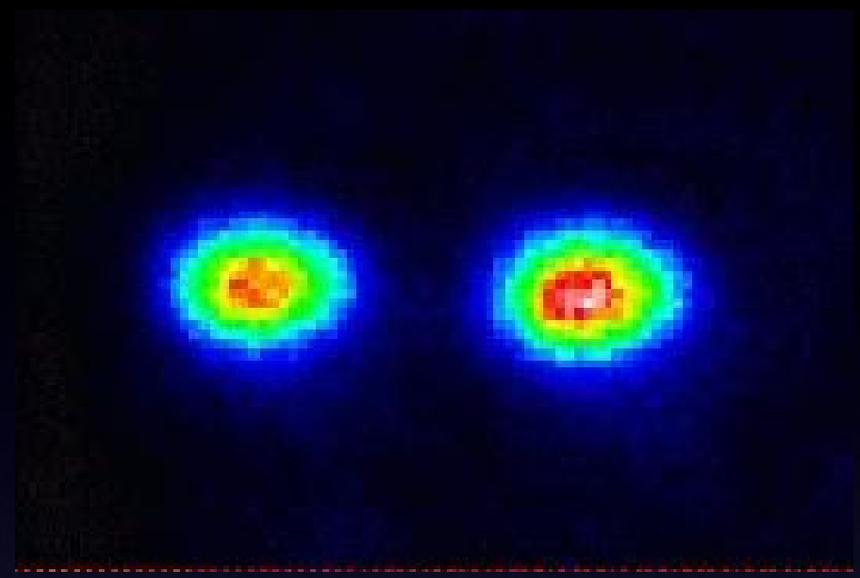




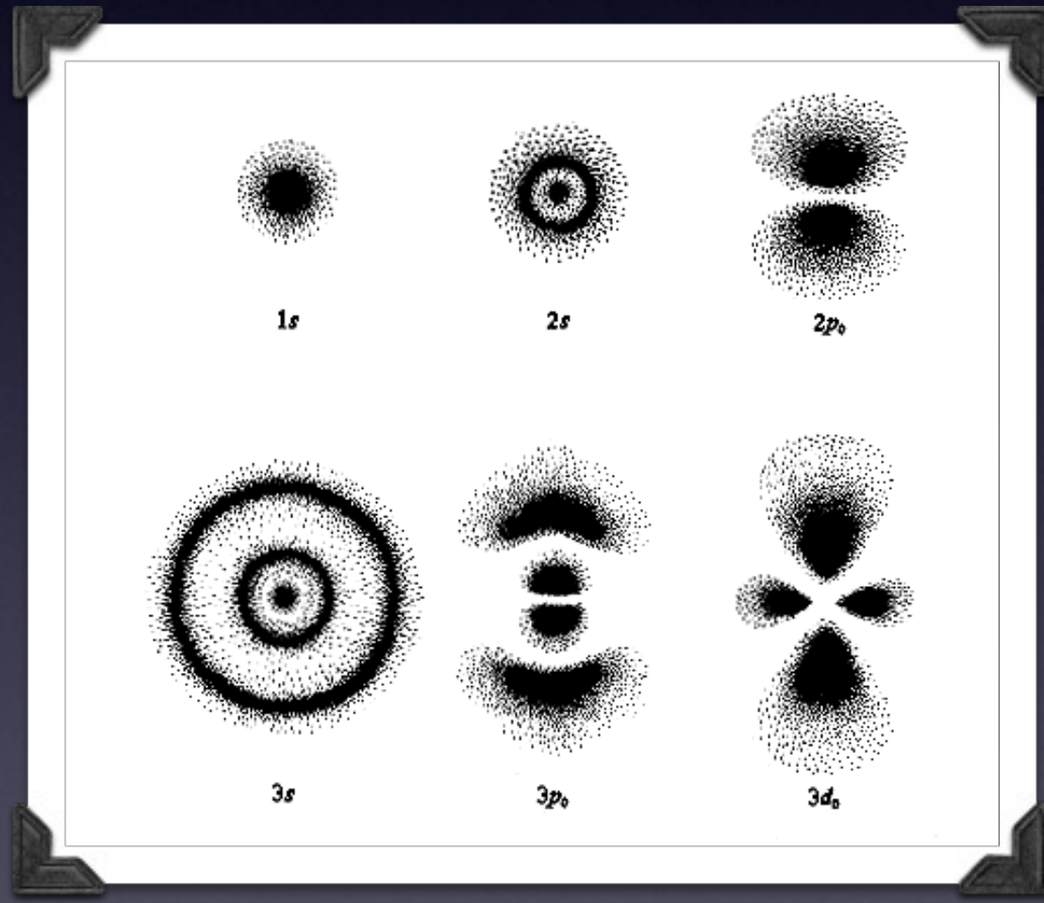
2 Be Ions in an Ion Trap

Where's the Qubit?





Each ion = 1 qubit

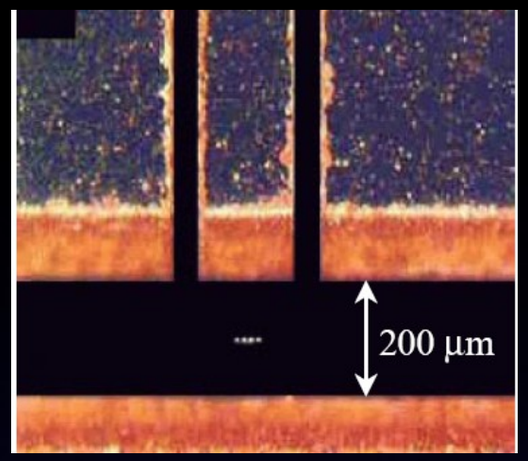


orbitals

Time Scales

walking

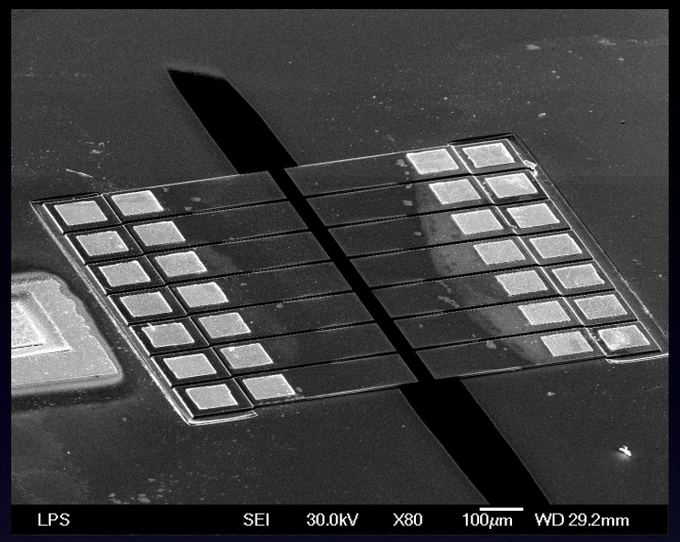
"slow"



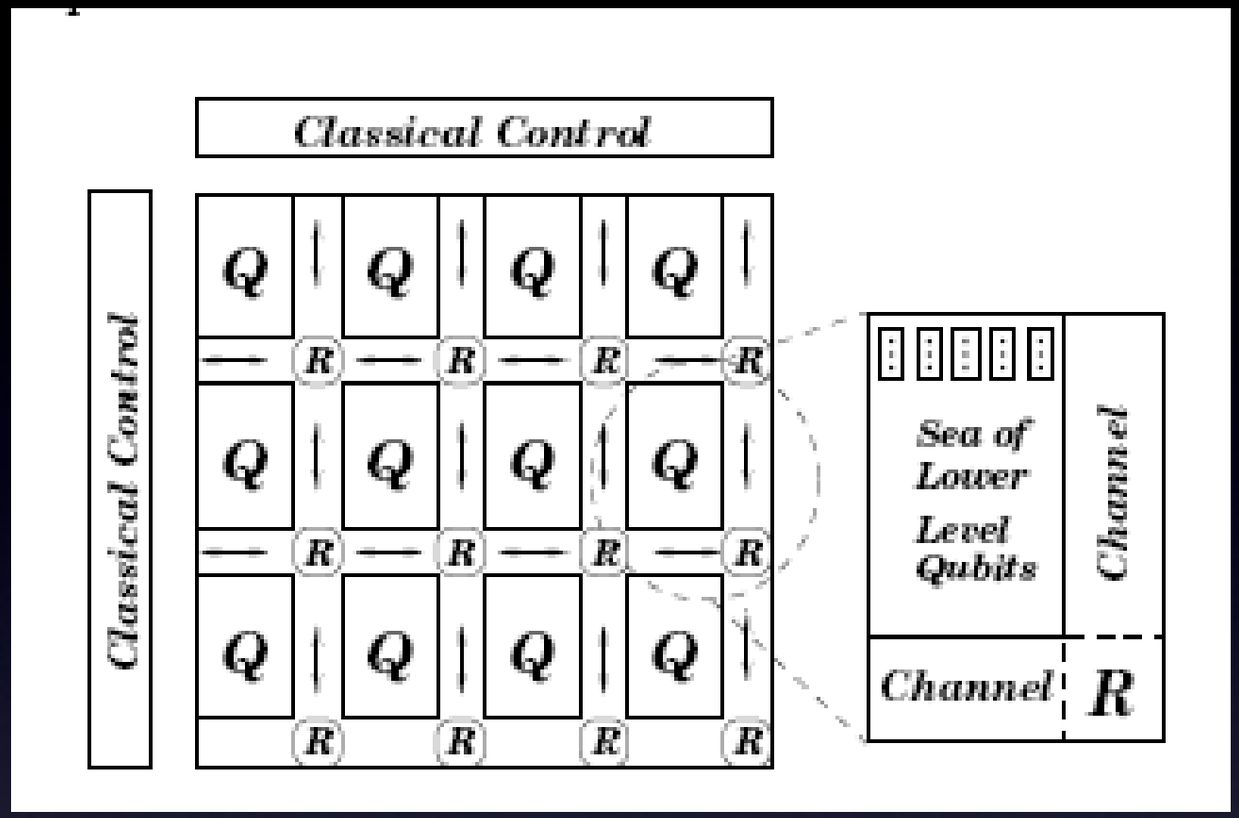
Example:
ion trap qubits

Operation	Time (□s)	Failure rate (today)	Failure rate (future)
single qubit gate	1	10^{-4}	10^{-8}
two qubit gate	10	10^{-2}	10^{-7}
measurement	200	10^{-2}	10^{-8}
movement	20	10^{-3}/ □m	10^{-7}/ □m
cooling	200	10^{-4}	?
wait	10^{-7} / error		

Sketch Ion Trap QC



http://iontrap.physics.lsa.umich.edu/research/trap



(from Metodi et al, MICRO-38)



Further Resources

Quantum Computing For Computer Scientists (David Mermin):
http://people.ccmr.cornell.edu/~mermin/qcomp/CS483.html

"Standard" Textbook:
Quantum Computation and Quantum Information
by Michael Nielsen and Isaac Chuang

ISCA 2002 Tutorial Page (Oskin and Chong):
http://www.cs.washington.edu/homes/oskin/quantum-tutorial/

My Lecture Notes on Quantum Computing:
http://www.cs.washington.edu/education/courses/cse599d/06wi/